

PROFESSIONAL SUMMARY

Army veteran and multidisciplinary engineer with a B.S. and M.S. in Chemical Engineering and ongoing M.S. studies in Computer Science. Tactical and strategic thinker with strong problem-solving ability, project management experience, and attention to detail. Skilled in software development, cloud technologies, applied research, process modeling, and controls with a proven ability to analyze complex systems and deliver reliable, high-quality technical solutions. Strong communicator who excels in fast-paced, mission-driven environments.

PROFESSIONAL EXPERIENCE

Emerson Automation Solutions | Austin, TX

Applied Research Engineer

January 2023 – Present

- Provided industrial subject-matter expertise to UT J.J. Pickle Research Campus, enabling the successful configuration and commissioning of their DCS system for autonomous control research initiatives
- Oversaw a six-member development team to design, build, and operationalize a comprehensive Azure Monitor-based observability solution, strengthening both operational insight and security visibility
- Contributed to the DevOps team responsible for deploying applications, provisioning cloud infrastructure, and maintaining a continuous delivery pipeline
- Coordinated a large-scale internship program, managing 100+ interns while developing and mentoring 10 interns directly

Software Development Engineer Graduate Intern

May 2022 – August 2022

- Implemented a prototype to allow embedded devices (Beagle Bone Board) to send sensory data to Azure IoT Hub (PaaS)
- Developed software using C/C++ and OOP to automate communication between sensors and embedded devices
- Created a dashboard using HTML5, CSS, and JavaScript to visualize data stored in the cloud

Software Development Engineer Graduate Intern

May 2021 – August 2021

- Programmed a reinforcement learning algorithm to self-tune a Temperature PID Controller (TCLab) using PyTorch infrastructure
- Researched different Machine Learning techniques to optimize Neural Network performance for supervised and reinforcement learning
- Awarded 2nd place in Emerson's Women in STEM Hackathon by implementing a supervised learning algorithm to anticipate run time in DeltaV architecture using Python

Invista | Victoria, TX

September 2022 – January 2023

Process Transformation Engineer

- Developed an anomaly detection script using an LSTM-Autoencoder algorithm using Seeq and Pytorch infrastructure
- Engineered a model to determine the decay rate of catalyst activity and heat exchanger fouling
- Designed a Plotly dashboard to help visualize advanced data analytics
- Incorporated forecasting algorithms using Prophet or LSTM to determine predictive maintenance

Solugen | Houston, TX

Oct 2019 – May 2020

Project Engineer

- Led the commission of Solugen's Slaton facility to receive 34% and 50% hydrogen peroxide by rail
- Managed the installation of a security system to meet Homeland regulation; executed a budget of \$25,000
- Constructed a Quality Assurance/Quality Control laboratory with a budget of \$40,000; ensured outgoing product specifications met customer expectations

- Coordinated the installation of a reverse osmosis unit to help dilute hydrogen peroxide; delivered the project to customer specifications and on with zero cost overruns on the \$15,000 budget

Gerdau Long Steel, North America | Midlothian, TX
Process Engineer Intern

Jun 2018 – Aug 2018

- Managed a \$50,000 project; oversaw the restoration of a slag analyzer and a spectrometer
- Sustained the reliability of 2 analytical instruments and decreased human traffic by 10%
- Revised lab procedure by implementing Six Sigma methodologies facilitated a 15% precision increase
- Prepared a preliminary risk assessment to minimize potential hazards, validated and secured the proper personal protective equipment, and obtained the appropriate permits in the production area

EDUCATION

Master of Computer Science | Rice University | 2027

Master of Science (M.S.), Chemical Engineering | Texas Tech University | 2022

Bachelor of Science (B.S.), Chemical Engineering | Texas Tech University | 2019

PUBLICATIONS

- Lakhani, A. I., Chowdhury, M. A., & Lu, Q. (2021). Stability-Preserving Automatic Tuning of PID Control with Reinforcement Learning. *arXiv e-prints*, arXiv-2112.